

# Harik Sodhi

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## AWARDS & HONOURS

- **Silver Medal, International Physics Olympiad** (2025) – 49th/420 competitors from 89 countries; highest-ranked competitor from Western Europe.
- **United Kingdom International Physics Olympiad Team Member** (2025) – Selected after Gold awards in BPhO Round 2 and BAAO Round 2.
- **Winner, Berkeley Math Tournament Online Power Round** (2024) – First place in a proof-based competition on graph theory, cryptography, and computational complexity.
- **Ritangle National Champion** (2023) – Team Captain; 1st/1,200 teams in a national combinatorial optimisation challenge, developing and proving an optimal solution to a highly constrained scheduling problem with hundreds of interacting variables.

## EDUCATION

**University of Oxford (Christ Church) | MEng Engineering Science** **Oct 2025 – Jun 2029**

- First-year: 93% labs, 92% internal exams

## EXPERIENCE

### Oxford Alpha Fund

**Oxford**

*Quant Analyst & Director*

*Jan 2026 – Present*

- Selected through a competitive quantitative research and presentation process as a first-year undergraduate.
- Researched the question "Do BTC options systematically misprice volatility, and does this vary by day of the week?"; developed and backtested systematic trading strategies achieving a Sharpe ratio of 1.18 and a 70% win rate with a short iron butterfly trade. [Code]
- Served on the general committee, contributing to the Walton Edge newsletter, sponsor liaison, and conducting interviews for joining the society.

### Optiver Trading Academy

**Oxford**

*Participant*

*Oct 2025 – Nov 2025*

- Selected for Optiver's competitive Trading Academy programme.
- Traded ETFs, futures, and options on Optiver's proprietary simulation platform Optibook, applying market-making, delta-hedging, and tuning parameters while competing against other teams.

## LEADERSHIP & ACTIVITIES

### VEX Robotics & REC Foundation

**London / Remote**

*Global Student Advisory Board Member & Team Captain*

*May 2021 – Mar 2024*

- Selected as one of 12 members globally to advise the CEO and executive leadership team, via monthly meetings, on the many different aspects of the REC Foundation—the governing body of the world's largest robotics competition spanning 2,400+ teams from 60+ countries and major corporate partnerships (e.g., NASA).
- Served as Team Captain, Lead Programmer, and Lead Designer for 2 years; managed team operations and helped develop the competition robot and strategy.
- Developed autonomous programs in C++ using PID control and designed competition mechanisms in CAD.

## PROJECTS

### Market Making Simulator

**Nov 2025**

- Developed a multi-agent market simulation modelling market makers, noise traders, and informed traders on a simple card game.
- Implemented strategy evaluation and performance analysis tools for comparing trading behaviour. [GitHub]

### Rocket Lander

**Apr 2026**

- Designed and developed a MATLAB simulation of autonomous rocket landing dynamics from first principles, incorporating variable-mass propulsion, atmospheric drag, and gravity.
- Implemented Runge-Kutta numerical integration, a proportional control system, robustness testing for variations of initial conditions, and constrained optimisation techniques to achieve stable autonomous touchdowns while reducing fuel consumption. [GitHub]

## SKILLS

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- **Programming:** Python, C++, MATLAB, LaTeX
- **Quantitative:** Statistical Modelling, Backtesting, Optimisation, Volatility Analysis, Numerical Methods
- **Libraries:** NumPy, Pandas, SciPy, PyTorch, Scikit-learn, Matplotlib
- **Tools:** Linux, Git, GitHub, SolidWorks (CSWE)